Architectural Simplification and Internal Software Quality in a Unity-Based Serious Game: A Comparative Case Study

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Abstract

Architectural refactoring is most often studied as the addition of structure. Removing abstractions that no longer carry a responsibility has received limited empirical attention, as has the reading of before-and-after metrics when the intervention itself changes the set of components measured. This paper reports a comparative case study of responsibility-driven architectural simplification in Molz, a Unity-based serious game. Three pinned states were measured with one frozen instrument: baseline C, a control state D0 obtained by removing dead code and nothing else, and the simplified state D, from eight staged interventions that retained an abstraction only where it carried a concrete, non-duplicated responsibility. The control mattered: the removed subset was unrepresentative, so deleting it alone moved every aggregate mean. Between D0 and D the system lost five classes and gained two interfaces and six members, while aggregate indicators moved little: maintainability index 71.7 to 71.8, cyclomatic complexity 2.39 to 2.38, approximate efferent coupling 12.8 to 13.4. A matched-population analysis over the 442 members present in both states reversed the direction of the first two, showing that composition, not the surviving code, set that direction. The set of engine-instantiated types was identical in both states. Behaviour covered by the validation protocol was preserved. We contribute evidence on which abstractions were removed and which retained, a dead-code checkpoint separating cleanup from design, and a population-aware reading of aggregate quality indicators.

Keywords: Software architecture, Architectural refactoring, Software quality, Software metrics, Game software engineering, Unity

1 Introduction

Refactoring is defined as a change to the internal structure of software that makes it easier to understand and cheaper to modify without altering its observable behaviour Fowler (2018); Mens and Tourwé (2004). In the empirical literature we surveyed, the activity was most readily found in one direction: structure is added. Classes are extracted, interfaces are introduced, responsibilities are distributed across new components. The catalogues of code smells contain the opposite case as well — Speculative Generality, Lazy Class, Middle Man, Dead Code Fowler (2018) — but among the studies we located, evidence for removing structure was less abundant than evidence for adding it, and scarcer still at the level of architecture rather than of individual classes. Our search was targeted rather than systematic (Section 2.5), so we report this as the shape of the evidence we found and not as a characterization of the field.

This matters because the two directions do not behave symmetrically under measurement. Aggregate indicators such as the maintainability index Coleman et al. (1994), cyclomatic complexity McCabe (1976) and approximate efferent coupling Martin (2003) are means over a population of members or types. When structure is added, the population grows; when structure is removed, the population shrinks. A mean can therefore move without any surviving component having changed at all. Studies that report only aggregate deltas cannot distinguish the two, and the literature already reports that refactoring produces mixed rather than uniform metric outcomes Szőke et al. (2017); Bavota et al. (2015); Palomba et al. (2018).

Game software adds a further dimension. Games are built on engines that impose their own composition model, lifecycle and serialization mechanics Aleem et al. (2016); Politowski et al. (2025), and engine-specific structures are a recognized source of maintainability problems Nardone et al. (2023); Borrelli et al. (2020). In an engine-based system, a component's dependency on the engine is not a free design variable: it is partly fixed by the way the engine instantiates and wires objects. Whether architectural simplification can reduce that dependency, and what happens to internal quality indicators when it is attempted, is an open empirical question.

This paper reports a comparative case study of *responsibility-driven architectural simplification* in Molz, a Unity-based serious game for the solubility classification of organic compounds. The guiding principle of the intervention was not to impose a target architectural style, but to reassess each existing abstraction against the responsibility it actually performed in the running system, and to retain only those carrying a concrete one.

The study is organized around three snapshots of the same system. State C is the pre-intervention baseline. State D0 is an intermediate control obtained by removing dead code and nothing else. State D is the result of eight architectural stages applied to D0. The control checkpoint was introduced after an audit found that a substantial

part of the baseline was unreachable; without it, the metric consequences of deleting unreachable code would have been credited to the architectural intervention.

Objective. To investigate how responsibility-driven architectural simplification of a functionally preserved Unity-based serious game changed its structural organization and internal software quality indicators, distinguishing the effect of dead-code removal from that of the subsequent architectural intervention.

RQ1. How did the system's architectural organization change when existing abstractions were reassessed according to their concrete responsibilities and redundant architectural elements were simplified?

RQ2. How did internal software quality indicators — maintainability, cyclomatic complexity, structural coupling, engine dependency, and structural size — differ between the pre-intervention state D0 and the simplified state D, and how do those differences relate to changes in component population?

RQ3. How were changes in internal software quality distributed across components removed as dead code, directly refactored, indirectly affected, structurally preserved, and newly introduced by the architectural intervention?

The contributions of the paper are four.

First, empirical evidence from a responsibility-driven architectural simplification carried out on a real Unity-based system under verified behavioural preservation, reported together with the decision rule, the staged protocol and the per-stage gates that produced it.

Second, evidence about *which* responsibilities and abstractions were removed or consolidated and which were retained, with the implementation evidence recorded for each — including two cases in which an a-priori simplification hypothesis was rejected by what the removal attempt revealed. Retentions are reported as outcomes, not as work left undone.

Third, a dead-code checkpoint separating population cleanup from the subsequent architectural intervention. In this case the separation mattered quantitatively: the removed subset was not representative of the system, so deleting it moved every aggregate mean on its own.

Fourth, a population-aware analysis showing how aggregate internal quality indicators can be affected by changes in the measured software population. For two indicators the direction of the aggregate movement was set by composition rather than by the code that persisted, although all magnitudes involved are small.

The remainder of the paper is organized as follows. Section 2 positions the study against work on internal quality measurement, architectural refactoring and game software architecture. Section 3 describes the case study design. Section 4 reports the results for the three research questions. Section 5 interprets them. Section 6 discusses threats to validity, and Section 7 concludes.

2 Background and Related Work

2.1 Internal Software Quality and Architecture

Maintainability is one of the product-quality characteristics of ISO/IEC 25010 International Organization for Standardization (2023), but the standard does not prescribe how to estimate it from source code. Practice has settled on a small set of structural proxies.

The *maintainability index* (MI) combines Halstead volume, cyclomatic complexity and lines of code into a single composite score Coleman et al. (1994). Its convenience is also its principal weakness: the composite was calibrated on a specific corpus, and the relationship between the score and the maintenance effort actually experienced by developers is indirect. Systematic reviews of maintainability prediction report wide variation in which metrics are used and how they are validated Riaz et al. (2009); Ardito et al. (2020), and the prediction line remains active, with recent work estimating maintainability risk from static code metrics with machine learning Şengül and Yürüm (2026). An influential alternative line of work argues that maintainability is better assessed by mapping ISO characteristics onto several separately reported source-code properties — volume, unit size, unit complexity, duplication, unit testing — rather than collapsing them into one number Heitlager et al. (2007), and by benchmarking those properties against a population of comparable systems Baggen et al. (2012). Static analysers are the usual vehicle for collecting such properties at scale, and recent longitudinal work confirms both their practicality and the care their output demands: Berciu et al. Berciu et al. (2026) analysed 3 656 releases of 57 projects and report quality-issue distributions that shift with language and with release history rather than holding as absolutes. In the same vein, Rantala et al. Rantala et al. (2024) found that keyword-labelled self-admitted technical debt and static-analysis findings — two common proxies for the same underlying quality problems — are related but overlap only to a limited extent, a reminder that what such instruments report depends on how the construct is operationalized.

Cyclomatic complexity McCabe (1976) has attracted a parallel critique. Shepperd Shepperd (1988) argued that the measure rests on weak theoretical foundations and that, for a large class of software, it is largely a proxy for size. Since MI incorporates both cyclomatic complexity and lines of code, the two critiques compound.

Coupling has a firmer conceptual basis. Chidamber and Kemerer Chidamber and Kemerer (1994) defined coupling between objects as the number of other classes a class depends upon, and Martin Martin (2003) defined efferent coupling at package level together with the dependency inversion principle, under which modules should depend on abstractions rather than on concrete implementations. Efferent coupling counts outgoing dependencies; it says how many other types a type must know about, not whether knowing about them is justified. Practitioners appear to reason the same way: in a questionnaire survey on recognising object-oriented design quality, Stevenson and Wood Stevenson and Wood (2018) found respondents placing fundamental principles such as low coupling above specific pattern vocabulary when judging a design. That is the concept; what our instrument computes is an approximation of it, obtained without symbol resolution, and we are explicit about the difference in Section 3.

We therefore treat MI, cyclomatic complexity and our coupling proxy as *internal software quality indicators* and not as measurements of maintainability itself. This distinction is not rhetorical: as Section 4 shows, two of these indicators move in one direction in aggregate and in the opposite direction once the population is held fixed.

2.2 Architectural Refactoring and Simplification

Surveys of refactoring Mens and Tourwé (2004), including its automated forms Baqais and Alshayeb (2020), and the catalogue of smells and corresponding transformations Fowler (2018) cover both the addition and the removal of structure. Empirically, however, the reported outcomes of refactoring on internal quality attributes are mixed rather than uniformly positive. Szőke et al. Szőke et al. (2017), studying large-scale industrial systems, found that a single refactoring had only a small and sometimes negative effect on maintainability, while the accumulated process had a beneficial one. Bavota et al. Bavota et al. (2015) found that refactoring operations are not a direct consequence of poor metric profiles or of the presence of smells. Palomba et al. Palomba et al. (2018) found that the impact of smells on maintainability varies substantially by smell type and by system.

One well-studied form of structural removal is dead and unused code. Romano et al. Romano et al. (2020) report a multi-study investigation showing that dead code is widespread and that developers perceive it as harmful to comprehensibility, and Eder et al. Eder et al. (2012) measured unused code in an industrial system and its relation to maintenance effort. This literature establishes that removing unreachable code is worthwhile; it does not report what removing it does to the aggregate metric profile of the system that remains — which is precisely the confound the control state D0 in this study is designed to isolate.

At the architectural level, the relevant body of work is architectural technical debt. Verdecchia et al. Verdecchia et al. (2020, 2021) built and evaluated a grounded theory of architectural technical debt, characterizing it as design decisions that were suitable when taken but later hinder progress — a description that fits abstractions whose justification has since disappeared. Kruchten et al. Kruchten et al. (2012) discuss the metaphor and its limits, and Li et al. Li et al. (2022) map the literature on architecture erosion and its symptoms. This work is largely diagnostic: it characterizes and detects degradation. Studies that apply a simplification intervention to a working system and then measure what happened, with a control for confounding deletions, are considerably rarer.

The transformation side of this literature is, meanwhile, increasingly LLM-assisted. Yang et al. Yang et al. (2026) automate best-practice refactoring in Java through multi-agent planning with explicit verification, observing that language models applied directly to refactoring tasks produce unsafe or incomplete edits unless they are placed under such control, and Gheventer et al. Gheventer et al. (2026) assess how ready generative-AI solutions for software quality actually are for industrial adoption. As interventions become cheaper to produce, the question of how their effects are measured — including the population confound examined here — becomes more pressing, not less; the intervention studied in this paper was itself implemented with LLM assistance under per-stage human approval (Section 3.5).

2.3 Game Software Architecture and Engine Dependency

Game development has been characterized as a distinct software engineering context, with its own process life cycle and its own recurring difficulties Aleem et al. (2016). Academic game software carries an additional one: Cerioli et al. Cerioli et al. (2026) argue that student-built serious games are routinely treated as throwaway artefacts and consequently decay into abandonware, and propose treating them instead as legacy software with a maintenance life cycle. Their framing matters for the present study, because it establishes that a serious game produced in an academic setting is a legitimate object of software maintenance research rather than only of game-design research. Engine architecture has recently received direct attention: Politowski et al. Politowski et al. (2025) recovered and compared the architectures of ten open-source engines, identifying excessive coupling as a recurrent issue and showing in a controlled experiment that explicit architectural models let developers perform understanding and impact-analysis tasks faster and more correctly. Ullmann et al. Ullmann et al. (2023) visualized subsystem coupling patterns across engines and found the low-level renderer, platform-independence layer and resource manager to be frequently coupled to the engine core.

Closer to the present case, Nardone et al. Nardone et al. (2023) derived a catalogue of 28 video-game bad smells from developer discussions, and Borrelli et al. Borrelli et al. (2020) implemented static detection of seven Unity-specific smells covering performance, maintainability and incorrect behaviour. Both establish that engine-specific structures are a legitimate object of internal quality analysis. Neither measures what happens to engine dependency when a game is deliberately restructured.

2.4 Measuring Architectural Change

The three strands above concern what to change and why. A fourth concerns how change is observed, and it is the one this study leans on most heavily.

Before-and-after measurement of a refactoring is the standard design, and the metrics used are typically those discussed above: size, complexity, coupling and a composite such as MI. What varies between studies is the unit over which those metrics are aggregated. Szőke et al. Szőke et al. (2017) report per-refactoring and per-process effects separately, which already acknowledges that the two answer different questions. Bavota et al. Bavota et al. (2015) relate metric profiles to the refactorings developers actually perform, rather than treating a metric movement as the effect of a known transformation. Longitudinal work such as Berciu et al. Berciu et al. (2026) tracks quality issues across releases, where the set of measured artefacts changes from release to release by construction.

A related concern is how per-component values are aggregated at all. Vasilescu et al. Vasilescu et al. (2011) show that the choice of aggregation technique over skewed software-metric distributions materially changes the conclusions drawn, and Mordal et al. Mordal et al. (2013) reach a similar verdict in industrial practice, designing dedicated aggregation schemes precisely because plain means over such distributions mislead. That literature varies the aggregation operator while the population of measured units stays fixed.

What we did not find discussed explicitly is the consequence for interpretation when the intervention itself adds and removes the units being averaged. An architectural change that deletes components, merges them or introduces new ones alters the population over which every mean is computed, so a before-and-after difference in that mean confounds two things: change inside the components that persist, and change in which components are counted. A study reporting only the aggregate cannot separate them, and nothing in the metric definitions signals that a separation is needed. The matched-population analysis in Section 4.2 exists to make that separation in this case.

2.5 Research Gap

Table 1 summarizes what the surveyed literature covers and what remains insufficiently addressed for the question of this paper.

The gap we claim is deliberately narrow. It is *not* that architectural simplification is unstudied: architectural technical debt research studies the conditions under which structure becomes a liability Verdecchia et al. (2020, 2021); Li et al. (2022), and refactoring research studies the removal of dispensable elements Fowler (2018); Romano et al. (2020). It is also not that mixed metric outcomes are unknown: they are reported repeatedly Szőke et al. (2017); Bavota et al. (2015); Palomba et al. (2018).

What our search did not find is the *combination* of three properties in a single study: (i) an intervention whose decision rule is the concrete responsibility of each existing abstraction, applied to a working engine-based system, with retention decisions recorded as results rather than as non-events; (ii) an explicit control state that separates the metric effect of deleting unreachable code from the metric effect of the architectural intervention; and (iii) an analysis that separates within-component change from population composition effects, so that aggregate movements are not read as changes in surviving code.

We state this as a gap in the evidence we located, not as a claim of priority. The literature search behind Table 1 was targeted rather than systematic: it followed the themes above through database and citation search until the returns became repetitive, and it was not conducted under a protocol that would license a statement that no such study exists. Each of the three properties has precedent individually; what this study offers is their combination, applied to an engine-based system and reported with the retention decisions intact.

Table 1 What the surveyed literature covers, and what remains insufficiently addressed for the question studied here.

Topic	Existing literature covers	Insufficiently addressed
Refactoring and metrics	Catalogues of transformations and smells; large-scale evidence that effects on internal quality are mixed and often small Fowler (2018); Mens and Tourwé (2004); Szőke et al. (2017); Bavota et al. (2015); Palomba et al. (2018)	Whether an observed aggregate movement reflects change in surviving components or a change in the population being averaged
Architecture and maintainability	Composite and multi-property maintainability models, their calibration and their limits Coleman et al. (1994); Heitlager et al. (2007); Baggen et al. (2012); Riaz et al. (2009); Ardito et al. (2020)	Behaviour of these indicators under an intervention that reduces the number of components rather than increasing it
Technical debt	Grounded theory of architectural technical debt; erosion symptoms and their detection Verdecchia et al. (2020, 2021); Kruchten et al. (2012); Li et al. (2022)	Applied interventions with before/after measurement and a control for confounding deletions
Game architecture and engine coupling	Engine architecture recovery and comparison; subsystem coupling patterns; Unity-specific smell catalogues and detectors Politowski et al. (2025); Ullmann et al. (2023); Nardone et al. (2023); Borrelli et al. (2020)	Whether restructuring a game changes the set of components that depend on the engine, as opposed to the proportion
Simplification and removal of abstractions	Dispensable smells; empirical study of dead code and unused code Fowler (2018); Romano et al. (2020); Eder et al. (2012)	Removal decided per abstraction by its concrete responsibility, with retention and rejected hypotheses reported as results

3 Case Study Design

The study follows the guidelines of Runeson and Höst Runeson and Höst (2009) for case study research in software engineering. In their classification it is an *improving* case study with an embedded design: the researchers intervened in the object and measured the outcome, and the analysis descends from the system as a whole to units within it. Following their recommended reporting structure, we describe in turn the case and its context, the objective and units of analysis, the states compared, the intervention and verification protocols, the measurement instrument, the component classification, and the analysis procedure.

3.1 Molz and Study Context

Molz is a mobile serious game for organic chemistry, implemented in Unity with C#. It represents the solubility classification procedure used to identify unknown organic compounds: the player performs virtual tests with solvents and reagents, observes simulated outcomes, and narrows the set of candidate compounds by elimination. The chemical responses are driven by a structured dataset rather than by a physical simulation, and the game offers three modes with decreasing levels of support.

The system was developed in an interdisciplinary academic context at the Universidade Federal do Agreste de Pernambuco, with contributions from computing, chemistry, education and design, and is publicly available as a research artefact.[1] Development followed Scrum with sprints tracked on GitHub Projects, and the architecture was documented as a layered Model-View-Presenter organization with Data, Domain, Core and Presentation layers. A user evaluation with 103 participants was conducted and reported separately; it is context for this study, not evidence within it.

This work sits inside a larger research programme. The umbrella project, *Software Process Improvement: Consolidating Evidence and Agile Practices*, investigates how software engineering methods and practices can be adapted and evaluated in real development settings; within it, the research plan *Investigation and Configuration of Methods and Processes for Quality Improvement in Digital Game Development* narrows that question to game software, and uses Molz as its empirical artefact. Under that plan the system had already been through a cycle of requirements work, agile process instrumentation, layered architectural restructuring, design-pattern adoption, functional verification and a first static-analysis campaign. **State C is the endpoint of that cycle**: it is not an untouched legacy baseline but a system that had already received deliberate engineering attention, which is why the audit reported in Section 4.1 found so many abstractions already carrying legitimate responsibilities, and why the intervention studied here became a reassessment rather than a migration.

An earlier 3D version of the game exists in the project's history. It is *not* a state of this experiment and is not used as a baseline or as a point of comparison. The entire experiment reported here takes place inside the 2D system.

Molz is a suitable object for this question for three reasons. It is small enough that every abstraction can be individually audited and every decision recorded, at 91 source files and roughly 5 000 lines of code in the baseline. It is real: it is a released

[1] https://github.com/bcccoworkingufape/game-quimica-2D

product with users, not a constructed example. And it is engine-based, so that engine dependency is a genuine design constraint rather than a stylistic preference.

3.2 Objective, Units of Analysis and Data Sources

Following the Goal-Question-Metric formulation Basili et al. (1994) used to make case study objectives explicit, the objective of this study is:

> *Analyse* a functionally preserved Unity-based serious game *for the purpose of* understanding how responsibility-driven architectural simplification changes structural organization and internal software quality indicators *with respect to* maintainability, cyclomatic complexity, structural coupling, engine dependency and structural size, *from the point of view of* researchers assessing architectural interventions, *in the context of* a single small engine-based system whose behaviour is held constant and in which dead-code removal is separated from architectural change by a control state.

Case. The case is the Molz 2D system — specifically its production source tree `Assets/Scripts` — observed at three pinned points in its history. It is a single case; the three states are states of that one case, not three separate cases.

Context. An academic serious-game project built on a commercial engine by a small interdisciplinary team, with the system already released and in use.

Units of analysis. The design is embedded: different research questions are answered at different units, and Table 2 records which. Conflating them would be a measurement error, because complexity and maintainability are member-level quantities in this instrument while coupling, size and engine dependency are type-level ones.

Table 2 Units of analysis and data sources per research question.

RQ	Unit of analysis	Data source
RQ1	Architectural concern (an abstraction or group of abstractions assessed as one decision)	Audit records, stage commits, retention rationales, source diff
RQ2	The system as a whole, plus matched types and matched members across two states	Per-type and per-member instrument records for D0 and D
RQ3	Source file, resolved to its types and members, grouped by intervention category	Instrument records joined to the category assignment derived from the commit history

Data sources. Four, used in combination: the frozen per-type and per-member measurement records for the three states; the version-control history, which fixes the states and yields the RQ3 classification; the intervention audit records, which hold each concern's evidence and decision including retentions; and the verification artefacts of the parity protocol. Only the first is quantitative.

3.3 Experimental States C, D0 and D

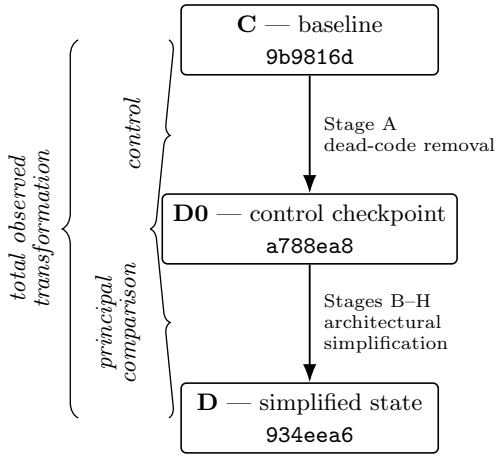


Fig. 1 Experimental design. The same system is observed at three states pinned to immutable commits. **D0 is a control checkpoint, not a third competing architecture**: it isolates the effect of deleting unreachable code from the effect of the architectural intervention, so that the principal comparison D0 → D is not confounded by it. States C and D also carry annotated tags, and each state has a frozen branch. The figure is methodological: no quality indicator appears in it.

Three snapshots of `Assets/Scripts` were defined, each pinned to an immutable commit and recovered for measurement in an isolated worktree (Figure 1). Each state additionally carries a frozen branch, and C and D carry annotated tags, so that a state can be recovered by commit, by tag or by branch and the three routes agree.

The three states support three comparisons:

- C → D0 — the effect of removing dead code (control);
- D0 → D — the effect of the architectural intervention (principal comparison);
- C → D — the total observed transformation.

D0 is not a third competing architecture. It exists because an audit of the baseline found that a substantial part of the Domain layer had no consumers. Deleting unreachable code changes every aggregate mean, because the deleted population is not a random sample of the system. Without D0, that change would have been indistinguishable from the effect of the architectural work.

3.4 Architectural Simplification Protocol

The intervention was not the application of a target architectural style. An initial hypothesis of migrating the system to Clean Architecture / Ports and Adapters was formulated, audited against the baseline, and *rejected before any code was written*: state C already had a Unity-free domain layer, separated Data, Domain and Presentation folders, and abstractions carrying concrete responsibilities. The decision rule adopted instead was:

For each existing abstraction, identify the responsibility it performs in the running system. Retain the abstraction if and only if that responsibility is concrete and is not already discharged elsewhere.

Four outcomes were possible for each abstraction: REMOVE, MERGE, STRENGTHEN (the abstraction is justified but bypassed in practice) and KEEP. KEEP decisions were recorded with their evidence, on the same footing as changes, so that a retention is reportable as a result rather than as the absence of one.

The intervention proceeded in eight stages (Table 3). Each stage was implemented in isolation, compiled, verified against the test suite, and approved by a human reviewer before the next stage began. Stage A alone produced D0; stages B through H produced D.

Table 3 Intervention stages. Stage A produced the control state D0; stages B–H produced the simplified state D.

Stage	Intervention
A	Dead-code removal: 11 files, 20 types
B	Data layer: one DTO and four mappers removed; transformation localized in the repositories
C	Audio boundary adopted: 56 concrete accesses to engine singletons reduced to 0
D	Partial explicit composition: three dead registrations removed; one presenter composed by constructor; service registry deliberately retained
E	Score multiplier consolidated into the domain scoring service
F	Indicator-reagent semantics given a single owner in the domain entity
G	View $\leftrightarrow$ orchestrator cycles broken through two role contracts
H	Panel-reset duplication replaced by the existing collaborator

Two constraints were imposed on the intervention and observed throughout. No `MonoBehaviour` was renamed, moved between namespaces, or relocated in a way that would invalidate an asset GUID, because that would silently break scene serialization. And no global access mechanism was to be replaced by another global access mechanism: a change that substitutes one ambient singleton for another was not counted as simplification.

3.5 Use of an LLM-Based Coding Assistant

The code changes constituting the intervention were implemented with the assistance of an LLM-based coding agent (Claude Code, Anthropic), operating inside the staged protocol described above: the agent proposed and applied changes within one stage at a time, and every stage was reviewed and explicitly approved by a human author, passed the functional-parity protocol described in the next subsection, and was subject to the recorded prohibitions before the next stage began. The responsibility assessments and

retention decisions reported in Section 4.1 were made and recorded under the same human gates. The measurement instrument and the analysis scripts were produced by the same process; every reported value is re-derived mechanically from the frozen per-type and per-member records by a versioned verification script, so no reported number rests on unreviewed assistant output. LLM-assisted refactoring is itself an emerging research subject Yang et al. (2026); here the assistant is part of the intervention procedure, not an object of study, and the study makes no claim about the effect of this assistance on productivity or on code quality. Section 6 discusses it as a threat.

3.6 Functional-Parity Protocol

The experimental target was behavioural preservation between C and D for the behaviour covered by the validation protocol. We do not claim, and did not attempt to establish, formal functional equivalence between the two states; what follows describes operationally what was verified and therefore the scope of the preservation claim. Because the object is an engine-based game whose wiring lives partly in serialized scene data, no single verification technique is sufficient, so five were combined.

Behavioural characterization tests. A baseline characterization suite of **68 tests** was written against state C *before* any modification, and covered data loading and deserialization, solubility rules, gameplay rules, session state, and the mapping behaviour of the data layer. That baseline was the reference for the dead-code control: state D0 passed the same 68 tests, unchanged.

The suite then grew *during* the intervention. Before or alongside each stage that touched a behaviour not yet pinned down, tests were added to characterize that behaviour first, so that the stage could be evaluated against it: a nine-case deserialization spike and two further data-loading cases before the DTO removal, repository materialization and entity invariant tests around the mapper consolidation, score multiplier tests before the scoring change, and indicator semantics tests before ownership of the reagent rule moved into the domain. Tests introduced at a gate were retained in force for every subsequent stage, unless the production abstraction they characterized had itself been removed. Exactly one such retirement occurred: the thirteen mapping tests went when the mapper types went, and their coverage was taken over by the repository materialization and entity invariant tests written for the arrangement that replaced them.

State D therefore ends with **118 tests passing**, of which **103 are behavioural characterization tests** and **15 are architecture invariant tests** (Table 4). The 118/118 figure is evidence about the final suite; it is not a claim that all 118 tests existed before the first change to C.

Architecture invariant tests. Fifteen of the final tests do not exercise behaviour at all. They scan the source of the measured tree and assert structural properties — that no call site outside the audio adapter reaches the engine audio singletons directly, and that the view contracts carry no engine types. These were introduced by the stages that established the properties they guard, and they are reported separately because they are evidence of *structural* conformance and contribute nothing to the argument for behavioural preservation.

The behavioural suite compiles the engine-free production files outside Unity; it is not part of the population measured by the instrument. It deliberately preserves several behaviours that would ordinarily be treated as defects, including a difficulty asset with a zero score multiplier, a divergence in the default difficulty, an inert branch in a UI controller, and the specific parsing policy applied to malformed dataset entries. Correcting any of them would have changed behaviour and broken the preservation target.

Table 4 Growth of the validation suite. The baseline was written against C before any modification; later tests were added by the stages that touched the behaviour or property they characterize.

Added	**Tests**	**Kind**
Baseline, against C	68	behavioural
unchanged through Stage A (D0)	*68*	*behavioural*
Deserialization spike, Stage B	11	behavioural
Repository materialization, Stage B	14	behavioural
Entity invariants, Stage B	5	behavioural
mapping tests retired with the mapper types	−13	*behavioural*
Audio boundary, Stage C	9	architecture invariant
Score multiplier, Stage E	8	behavioural
Indicator semantics, Stage F	10	behavioural
View boundary, Stage G	6	architecture invariant
State D, final	**118**	**103 + 15**

Compilation. Both through an isolated project definition and through the engine's own assembly, in each state.

Engine import. The Unity project was imported in state D with zero missing-script errors and zero serialization errors, confirming that no GUID or serialized binding was invalidated.

Manual full-flow verification. A scripted walkthrough of the complete game flow in all three modes.

Smoke verification after the final stage, covering phase restart, panel reset and return to menu.

3.7 Measurement Instrument

All three states were measured with a single instrument, frozen at one commit and never read from inside the measured worktrees. It consists of a Roslyn-based metrics extractor built on `Microsoft.CodeAnalysis.CSharp` 4.11.0, plus SonarAnalyzer.CSharp 10.31.0.145097 and the engine's own analyzers, executed with the `dotnet` SDK 9.0.306 under Unity 6000.4.1f1. Each state is recovered with a detached worktree at its pinned commit, so production content and measurement tooling remain separate; no snapshot is modified to accommodate the instrument.

The instrument reports, per type, lines of code, an approximate efferent coupling count defined below, and whether the type is engine-instantiated; and, per member, cyclomatic complexity and maintainability index. Aggregations are computed from the raw per-type and per-member records rather than from summary output.

Maintainability index. The instrument computes the Microsoft variant of the maintainability index Coleman et al. (1994): $\mathrm{MI} = \max\big(0,\ \frac{100}{171}(171 - 5.2\ln V - 0.23\,\mathrm{CC} - 16.2\ln \mathrm{LOC})\big)$, where V is Halstead volume — clamped at zero, rescaled to a 0–100 range, and without the comment-percentage term of the original formulation. It is computed per member and aggregated as a mean.

Engine dependency. A type is counted as engine-dependent when its declared base list contains `MonoBehaviour` or `ScriptableObject` — that is, when the type is one the engine itself instantiates, serializes and drives through its lifecycle. The check is syntactic and considers only the *directly* declared base types.

Approximate efferent coupling. For each type the instrument counts the distinct capitalised identifiers appearing anywhere inside the type declaration. This approximates Martin's efferent coupling Martin (2003) without resolving symbols, and we therefore report it as a *structural coupling proxy* rather than as C_e itself. It over-counts, because a capitalised identifier also matches an invoked method name, a property, an enum member or a constant; and it under-counts dependencies expressed without a capitalised identifier. Section 6 states the consequences. We use it because it is stable across the three states and computed identically for all of them, so differences remain comparable even though the absolute level is not a count of depended-upon types.

Two properties of the instrument were verified explicitly. *Determinism*: re-executing the pipeline on state D produced byte-identical artefacts, including the analyzer SARIF. *Version equivalence*: an error in the project-generation step was found and corrected during the study; after the correction, state C was re-measured through the real entry point and the resulting files were byte-identical to those produced before, so all three states are measured by the same instrument version.

Analyzer violation counts were also collected (C 197, D0 191, D 143). They are auxiliary pipeline output and are not used to answer any research question here; interpreting individual findings belongs to a separate study.

3.8 Component Classification

To answer RQ3, every source file was assigned to exactly one category. The assignment was derived mechanically from the commit history — which files each stage touched — rather than from post-hoc judgement.

DEAD CODE REMOVED Present in C, removed exclusively in C → D0, after triple evidence of unreachability: no C# references, no GUID references in scenes, prefabs or assets, and no occurrence of the type name in serialized YAML.

DIRECTLY REFACTORED Component whose code or responsibility was a deliberate target of stages B– H.

INDIRECTLY AFFECTED Component changed only to accommodate a change whose primary target was another component.

PRESERVED Existing component deliberately left unmodified by the relevant architectural interventions.

NEW INTERVENTION ARTIFACT Element absent from the baseline, introduced as a necessary part of the intervention.

A component that does not exist in a given state is recorded as *absent*, never as zero. A NEW INTERVENTION ARTIFACT therefore has no row for C or D0, and DEAD CODE REMOVED has no row for D0 or D. Substituting zero would inject artificial values into every mean.

These five labels map onto the four population roles that matter for the analysis in Section 4.2: components *removed* from the measured population (DEAD CODE REMOVED), components *added* to it (NEW INTERVENTION ARTIFACT), components *modified* while remaining in it (DIRECTLY REFACTORED and INDIRECTLY AFFECTED, kept apart because the first were deliberate targets and the second were not), and components *preserved* unchanged (PRESERVED). We use the five-label form throughout, because it is the form recorded in the analysis scripts and in the replication package; renaming it in the manuscript alone would break that correspondence.

3.9 Analysis Procedure

Three analyses were performed on the frozen measurements.

Aggregate comparison. All summary values were recomputed from the raw per-type and per-member records in each state and cross-checked against the instrument's own summary. All 48 values — 16 indicators across three states — matched.

Matched-population analysis. Aggregate means confound two phenomena: change inside surviving components, and change in the population being averaged. Components were therefore matched across states by a stable identity key — `file|type` for types and `file|type|member` for members — and the mean was recomputed over the intersection only. Keys that were not unique within a state, which occurred for two method overloads in each state, were *excluded* from matching rather than resolved by assumption.

Category analysis. Metrics were aggregated within each RQ3 category, with units kept explicit: lines of code, approximate efferent coupling and type counts are per-type quantities; cyclomatic complexity and maintainability index are per-member quantities, and group-level values for them are means over the members belonging to the group, never aggregated per type.

Inferential significance testing was not used, because the study observes the complete measured component population of a single system at fixed states and does not posit a sampling model that would justify such inference. All reported quantities are descriptive, and no test is reported.

Chain of evidence. Every quantitative statement in Section 4 can be traced backwards through a fixed path: from the reported value, to the aggregation script, to the per-type or per-member record in the frozen result set, to the state's pinned commit, to the source file it was computed from. Every architectural statement traces from the claim, to the row of Table 5, to the audit record holding that concern's evidence and

decision, to the stage commit that implemented it — or, for a retention, to the recorded rationale for not changing anything. The RQ3 category of a file is determined by which stage commits touched it, whereas its metrics are computed from the measured source states, so the two are read from *distinct* artefacts. They are not independent in a statistical or causal sense: both ultimately derive from the same source changes, which is why Section 4.3 states the circularity in the containment result rather than treating the classification as externally validated.

4 Results

4.1 RQ1 — Architectural Reorganization

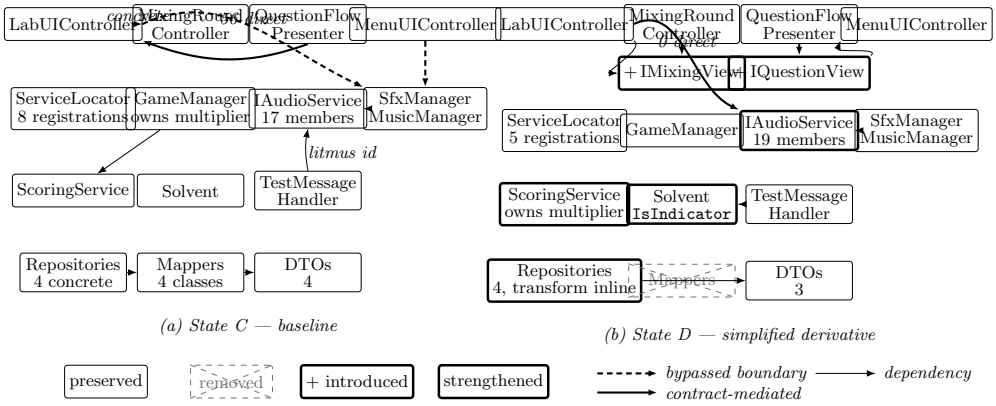


Fig. 2 Architectural component and dependency view of the two states, drawn at the same level of abstraction and with preserved components in the same positions so that the panels can be read against each other. State D is a simplified derivative of the baseline, not a different architectural family: most components are preserved, four mapper classes and one DTO are removed, two role contracts are introduced, and three boundaries are strengthened. Encoding does not rely on colour: preserved components have a plain border, removed ones a dashed border with a cross, introduced ones a heavy border with a +, and strengthened ones a heavy border. Every element and edge shown was verified against the pinned snapshots.

Figure 2 shows what changed structurally between the two states, and Table 5 records the reasoning behind each change and each retention. The figure answers *what* moved; the table answers *why*.

Table 5 records the outcome of the responsibility assessment for each abstraction examined. Thirteen concerns were audited: three led to removal or merging, two to strengthening, three to a local simplification or a change of ownership, and five to a documented retention.

Table 5: RQ1 — responsibility assessment of each abstraction. Retention decisions are reported with the same status as changes.

Concern	Initial hypothesis	Evidence found	Decision	Consequence
`QuestionMapper`	Redundant	Purely positional pass-through; invariants live in the entity	REMOVE	Binding now rests on constructor parameter names
`QuestionDto`	Redundant	1:1 copy, no type conversion	REMOVE	Verified by a nine-criterion spike before removal
`CompoundMapper`, `SolventMapper`, `SolutionMapper`	Duplicated	One caller each; real conversion present	MERGE	Transformation became a private repository method
Remaining three DTOs	Useful	Case conversion, string-to-enum, one rename	KEEP	Representation boundary preserved
`Json*Repository` public surface	Cleanup candidate	Four methods without consumers, three of which lost them *in Stage A*	KEEP	Removing them would credit the intervention with a control effect
`IAudioService`	Necessary but bypassed	56 concrete accesses against 2 port consumers	STRENGTHEN	56 → 0; port gained two availability members
`ServiceLocator`	**Redundant**	Full removal would require scene scanning, layer inversion or a lifetime change	**HYPOTHESIS REJECTED** → KEEP + SIMPLIFY	Three dead registrations removed; registry retained for cross-scene lifetime
`GameManager` responsibilities	Broad decomposition	Only scoring was split across layers	Selective SIMPLIFY	One move, seven documented retentions
Litmus indicator semantics	Duplicated in Presentation	Confirmed at two sites	MOVE	Ownership moved to `Solvent.IsIndicator`
`TreeSolventGroupButtons`	Duplicated	The consumer sustaining the duplication was removed *in Stage A*	**HYPOTHESIS REJECTED** → KEEP	Content is editorial and carries markup
View ↔ orchestrator cycles	Breakable	Two concrete reverse dependencies, two engine-type leaks	STRENGTHEN	`IMixingView` and `IQuestionView` introduced; `ILabView` untouched
God-object candidates	Decomposition	Only one duplication had a better existing owner	Local SIMPLIFY	One consolidation, nine retentions
Pre-existing dead members	Dead	Already dead in C	PRESERVE FOR PARITY	Retained so control and intervention stay separable

Simplification did not mean minimising abstractions. Structurally, the system that followed the intervention had *fewer classes and more interfaces*: from D0 to D the class count fell from 65 to 60, the interface count rose from 12 to 14, and the member count rose from 451 to 457. Five files were removed and two were added.

This is the opposite of the shape usually associated with simplification, and it is a direct consequence of the decision rule rather than an anomaly. The rule was not "reduce the number of abstractions"; it was "retain an abstraction exactly when it carries a concrete responsibility". Applied consistently, that rule moves in both directions at once, and the four decisions that most clearly show it are:

- `QuestionMapper` — **removed**: it performed a positional copy the entity constructor already performed.
- `IAudioService` — **strengthened**: the responsibility was real, but 56 call sites bypassed it; the port gained two members and the bypasses went to zero.
- `ServiceLocator` — **retained and simplified**: the removal attempt revealed a cross-scene lifetime responsibility nothing else discharged; three dead registrations were removed and the registry kept.
- `IMixingView` and `IQuestionView` — **introduced**: a responsibility existed — decoupling two views from their orchestrators — that no abstraction expressed.

We therefore do not report abstraction count as a measure of simplification, and a reader should not infer from the interface count that the system became more elaborate. What changed is which abstractions exist, not how many.

Four mechanisms of abstraction utility. Looked at again, those four decisions are not four instances of one rule. They are four different reasons an abstraction can or cannot be justified, and the difference matters for how a candidate should be assessed:

1. *No current responsibility* (`QuestionMapper`) — removed. Decidable by reading the source.
2. *A concrete responsibility, bypassed in practice* (`IAudioService`) — strengthened. Also decidable by reading the source, provided one counts call sites rather than inspecting the abstraction alone.
3. *A responsibility only the removal attempt revealed* (`ServiceLocator`) — retained. *Not* decidable by reading the source: the responsibility lived in the runtime composition model, which for an engine-based system is partly held in serialized scene data.
4. *A justification an earlier stage had already removed* (`TreeSolventGroupButtons`) — retained. Decidable when first assessed, and wrong by the time it was acted on.

Cases 3 and 4 both end in a retained abstraction, and it would be convenient to file both as false positives of the audit. That would lose the distinction that matters: in case 3 the analysis was incomplete, whereas in case 4 the analysis was correct and the system moved underneath it. The first argues for attempting a removal before trusting a static verdict; the second argues for re-deriving the candidate list after each stage rather than fixing it at the outset.

4.2 RQ2 — Internal Quality Indicators

4.2.1 Aggregate measurements

Table 6 reports the full aggregate profile of the three states.

Table 6 Aggregate measurements across the three states. CC and MI are per-member quantities; LOC, coupling and type counts are per-type.

Indicator	C	D0	D
Source files	91	80	77
LOC (non-blank)	5 093	4 738	4 729
Comment lines	695	649	825
Classes	79	65	60
Interfaces	16	12	14
Structs / records	7	5	5
Enums	9	9	9
Members	496	451	457
CC mean	2.33	2.39	2.38
CC median	2.0	2.0	2.0
CC max	17	17	17
CC total	1 158	1 078	1 089
MI mean	72.5	71.7	71.8
MI median	71.2	70.2	70.9
Members with MI ≥ 65	323	287	293
as % of members	65.1	63.6	64.1
Members with CC > 10	6	6	5
Approx. efferent coupling	11.4	12.8	13.4
Engine-coupled classes	24	23	23
as % of classes	30.4	35.4	38.3

Between D0 and D the system lost 3 files, 9 lines of code and 5 classes, and gained 2 interfaces and 6 members. The three principal indicators moved very little: MI mean rose by 0.1 on a 0–100 scale, CC mean fell by 0.01 on a scale whose observed mean is about 2.4, and the mean coupling proxy rose by 0.6.

Between C and D0 — the control — MI mean fell by 0.8 and mean efferent coupling rose by 1.4, even though the only action taken was the deletion of unreachable code. Figure 3 plots the three principal indicators across the three states: in each panel the control step moves the mean more than the intervention step does.

4.2.2 Matched-population analysis

The aggregate figures above mix two phenomena: change inside components that exist in both states, and change in the population over which the mean is taken. Table 7 separates them.

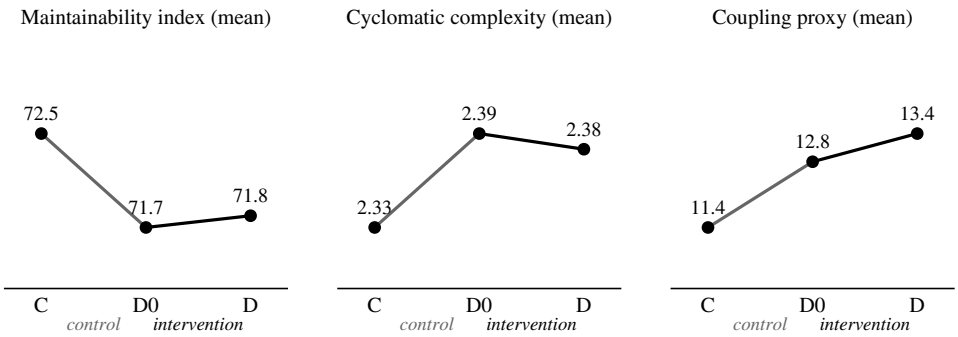


Fig. 3 Aggregate trajectories of the three principal indicators across the states (values from Table 6). The C → D0 segment is the control — dead-code removal only — and the D0 → D segment is the architectural intervention. The control step moves every mean more than the intervention step.

Table 7 Matched-population analysis, D0 → D. Holding the population fixed reverses the direction of the system-wide MI and CC movement; the matched members themselves also change, by small amounts.

Basis	**D0**	**D**	**Δ**
Members, system-wide (n = 442 matched)			
MI, aggregate mean	71.7	71.8	+0.1
MI, matched members	71.87	71.80	−0.06
CC, aggregate mean	2.39	2.38	−0.01
CC, matched members	2.41	2.42	+0.01
Directly refactored types			
Coupling, matched (n = 19)	26.84	28.37	+1.53
Coupling, raw group mean	22.29	28.37	+6.08
Coupling of removed types	5.00	—	—

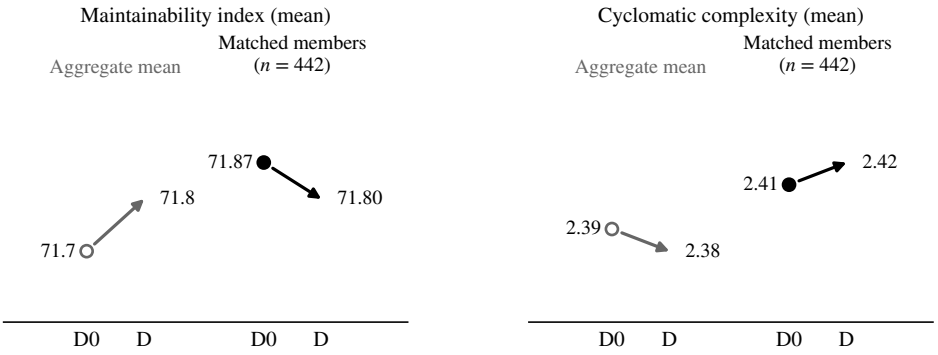


Fig. 4 Aggregate versus matched-population movement of the two per-member indicators, D0 → D (values as printed in Table 7). Holding the population fixed (black) reverses the direction that the aggregate mean (grey) reports for both indicators.

Three observations follow.

First, the direction reverses (Figure 4). The aggregate MI mean rises while matched-member MI falls; the aggregate CC mean falls while matched-member CC rises. Two phenomena are present at once, and neither is absent. Population composition — the eleven new members, largely simple properties scoring high on MI, and the five removed ones — reversed the direction of the aggregate signal. At the same time the surviving members did change, by −0.06 on MI and +0.01 on CC. The correct statement is therefore that composition reversed the direction of the aggregate movement, while the change among surviving members was small and opposite in direction to it — not that the surviving code was unchanged. Reporting the aggregate delta alone would have supported the statement that maintainability improved slightly; the matched analysis does not support that reading.

Second, the magnitudes are small in absolute terms. Only 65 of 442 matched members (14.7%) changed MI at all, and only 12 (2.7%) changed CC. The movements are 0.06 on a 0–100 scale and 0.01 on a scale whose observed mean is about 2.4. We do not infer practical significance from changes of this magnitude, in either direction, and we did not adopt a threshold that would license such an inference.

Third, the coupling increase is real and partly genuine. Of the +6.08 raw increase in the directly refactored group, +1.53 is within-type change among the 19 surviving types, and the remainder follows from removing five types whose mean coupling (5.00) was far below the group mean. Fourteen of the 19 survivors changed, twelve upwards and two downwards; the largest increases were in the repositories that absorbed the mapper logic — one rose from 12 to 19, another from 14 to 19, another from 12 to 15. Architectural responsibility redistribution and measured coupling increase are related here, but they are not the same phenomenon, and the increase is not merely a redistribution artefact.

4.2.3 Engine dependency

The set of engine-instantiated types — those directly declaring `MonoBehaviour` or `ScriptableObject` as a base type — was compared by identity, not by count:

Engine-coupled types in D0	23
Engine-coupled types in D	23
Intersection	23
Present only in D0	— (empty)
Present only in D	— (empty)

The set is identical. The percentage rose from 35.4% to 38.3% because the class population fell from 65 to 60 while the absolute count stayed constant. The percentage remains an accurate description of each state's composition; what it does not indicate is that any type entered the engine-coupled set, since that set was the same in D0 and D.

This is the clearest single result of the study, and it contradicted an expectation that had been central to the earlier framing of the work. Section 5 examines what this indicator does and does not capture. Here we report only that the set of types

directly extending `MonoBehaviour` or `ScriptableObject` was the same in both states, and that the movement of the percentage is accounted for by the change in the class population.

4.3 RQ3 — Distribution Across Component Categories

Table 8 gives the distribution of files across the five categories in each state, and Table 9 the group metrics.

Table 8 RQ3 — distribution of source files by category. A dash denotes absence, not zero.

Category	C	D0	D
Directly refactored	24	24	19
Indirectly affected	1	1	1
Preserved	55	55	55
Dead code removed	11	—	—
New intervention artifact	—	—	2
Total	**91**	**80**	**77**

Table 9 RQ3 — group metrics. The upper block is the intervention (D0 → D); the lower block is the control (C → D0).

Group	Types	Members	LOC	CC mean	MI mean	Coupling mean	Engine
Directly refactored, D0	24	199	2 357	2.44	71.2	22.29	8
Directly refactored, D	19	204	2 505	2.43	71.4	28.37	8
Indirectly affected, D0	3	11	58	1.36	83.5	7.00	0
Indirectly affected, D	3	12	67	1.33	84.0	7.67	0
Preserved, D0	**64**	**241**	**3 286**	**2.40**	**71.5**	**9.56**	**15**
Preserved, D	**64**	**241**	**3 286**	**2.40**	**71.5**	**9.56**	**15**
New intervention artifact, D	2	0	30	n/a	n/a	0.50	0
Control, C → D0							
Dead code removed (C only)	20	45	356	**1.78**	**80.3**	**4.50**	—
Whole project in C	111	496	—	2.33	72.5	11.4	—

The new intervention artifacts have zero members because both are pure interfaces; the instrument assigns no cyclomatic complexity or maintainability index to declarations without bodies. Their coupling of 0.50 is reported as measured.

The preserved group is metric-identical. Across D0 → D, the 64 preserved types show no change in any reported indicator: same type count, same member count, same lines of code, same complexity, same maintainability, same coupling, same engine-coupled count. In the remaining groups the control transition C → D0 is likewise near

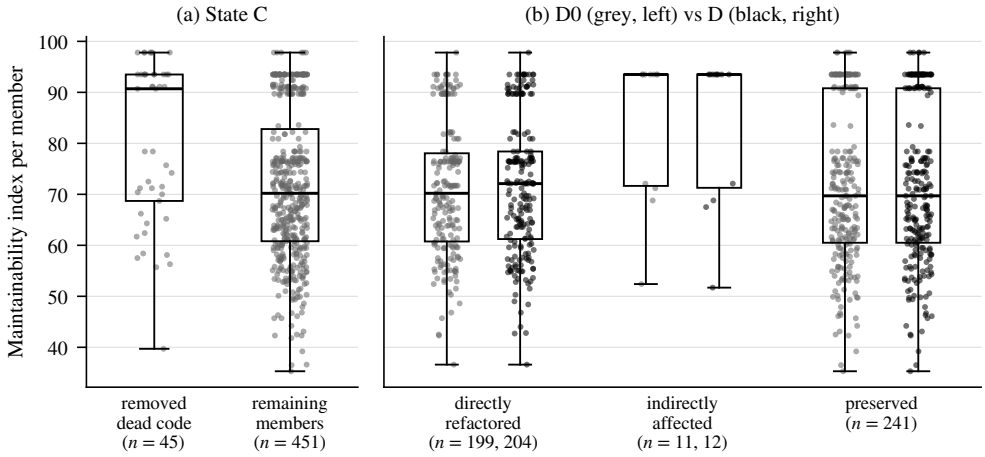


Fig. 5 Per-member maintainability-index distributions underlying Table 9 (boxes: quartiles and median; points: individual members). Panel (a): in state C, the members of the code removed as dead sit visibly above the rest of the system. Panel (b): per category, D0 (grey, left) against D (black, right); the preserved group is identical by construction, and the directly refactored group changes only slightly. The new intervention artifacts contribute no measured members and are omitted.

zero: −7 lines and −0.12 coupling in the directly refactored group, −0.02 coupling in the preserved group, and no change at all in the indirectly affected group.

The removed dead code was not representative (Figure 5a). The 20 types deleted in the control had mean MI 80.3 against 72.5 for the project, mean cyclomatic complexity 1.78 against 2.33, and a mean coupling proxy of 4.50 against 11.4. They were smaller, less complex and less coupled than the system average — which is unsurprising, since code that nothing calls tends to be small and to depend on little. Removing them therefore changed the composition of the population over which every mean is taken, and each aggregate mean moved in the less favourable direction as a result. That movement is a property of the subset removed, not a statement about the quality of the code that remained. This is the confound the control state was introduced to isolate, and it is quantitatively large enough to have swamped the effect of the architectural intervention had the two been measured together.

Interpretation and its limit. The measured structural changes fall outside the source-preserved group, which remained metric-identical, and inside the groups classified as directly or indirectly affected. This is evidence that the changes were *contained*. It is not an independent validation of the classification and not a causal attribution: the preserved group is *defined* by the absence of relevant source changes, and the metrics are derived from static code, so its metric identity partly follows from its definition. Classification and result are not independent, and we do not claim they are.

4.4 Summary of Unexpected Findings

Six results ran against the expectations held at the start of the study.

1. Removing dead code changed the composition of the measured population and moved every normalized aggregate mean in the less favourable direction, because the removed subset scored above the system average on every indicator.
2. Engine dependency did not fall: the same 23 types in D0 and D, with the percentage rising through denominator change alone.
3. Simplification was accompanied by fewer classes but *more* interfaces and more members.
4. Two a-priori simplification hypotheses were rejected by implementation evidence, for two different reasons.
5. The MI and CC movements reverse direction under matched-population analysis, although all magnitudes are small.
6. Coupling rose in the repositories that absorbed the mapper logic, both by redistribution and as a genuine within-type increase.

5 Discussion

5.1 Aggregate Indicators and Population Change

The clearest methodological lesson of this study is that an aggregate mean over a population that is itself being modified is a poor instrument for evaluating an intervention that modifies the population.

Two of our three principal indicators change direction between the aggregate and the matched view (Table 7). Had we reported only the aggregate, the natural narrative would have been that simplification produced a small improvement in maintainability and a small reduction in complexity. The matched analysis shows something different on two counts at once: within the 442 members present in both states, maintainability moved down by 0.06 and complexity up by 0.01, and the aggregate signal was pulled the other way by eleven newly introduced members, mostly simple properties that score well on MI. The surviving code did change; what the composition of the population did was reverse the direction the aggregate reported.

Both readings are arithmetically correct. They answer different questions. The aggregate answers "what is the profile of the system that now exists"; the matched analysis answers "what happened to the code that survived". For evaluating an intervention, the second is the relevant one, and it is the one the literature reports less often. The aggregation literature has long warned that the operator chosen over skewed metric distributions can change the conclusion Vasilescu et al. (2011); Mordal et al. (2013); the population effect observed here is the complementary hazard, produced not by the operator but by the intervention silently redefining the set being operated on.

The same point applies with more force to the control transition, and there it applies cleanly. Removing unreachable code is uncontroversially good practice Romano et al. (2020); Eder et al. (2012), and it moved every normalized indicator in this system in the less favourable direction. In that transition no surviving code was edited at all, so the entire movement is attributable to the change in the population being averaged. A study that combined dead-code removal with an architectural intervention and reported the aggregate delta would have attributed this movement to the architecture.

This is consistent with, and gives a concrete mechanism for, the repeatedly reported finding that refactoring produces mixed rather than uniform metric outcomes Szőke et al. (2017); Bavota et al. (2015); Palomba et al. (2018). Part of the mixture may be population effects rather than disagreement about the value of the transformations.

5.2 What the Engine-Dependency Indicator Captures

The engine-coupled set was identical in the D0 and D states: the same 23 types, with empty differences in both directions. This is the result that most directly contradicted the study's initial framing.

Interpreting this result requires being precise about what the indicator measures. Our engine-dependency indicator counts types that directly declare `MonoBehaviour` or `ScriptableObject` as a base type: the types the engine itself instantiates and drives. That is one operational manifestation of engine dependence, not a complete characterization of it. A type can use `UnityEngine` APIs extensively — for vectors, coroutines, resource loading or scene management — without extending either base class, and such a type is not counted here. In state C, thirteen types reference `UnityEngine` without being engine-instantiated; in state D, twelve. A system can likewise depend on an engine through its lifecycle assumptions, its serialization model or its scene composition without that dependence appearing in this count at all. Engine dependence is architecturally consequential — the engine architecture literature treats coupling to the engine core as a first-order concern Politowski et al. (2025); Ullmann et al. (2023) — and the point here is not that it carries no architectural meaning, but that our indicator captures only its explicit structural facet and should not be read as a measure of architectural decoupling.

Within that facet, the reason the set did not move follows from how the engine composes an application. A component a scene instantiates must be a `MonoBehaviour`; a field the editor serializes must be visible to the engine's serializer; a lifetime spanning scene loads must be managed by an engine mechanism Unity Technologies (2025). Our interventions redistributed responsibilities among the layers above these obligations, and none of them changed which components the engine must itself instantiate and wire. A different intervention — extracting engine-facing logic into engine-free collaborators, or changing the scene composition — might have moved the set. Ours did not attempt that, and this study cannot say what such an intervention would have produced.

The reporting implication is narrower and firmer. Because the percentage form of the indicator takes the class population as its denominator, it moves whenever that population moves. Reading 35.4% → 38.3% as increased engine dependence would be a denominator artefact, and reading a hypothetical downward movement as reduced dependence would be equally unfounded. Where the population changes, the set and its differences carry the information; the percentage does not.

The reporting practice this suggests is correspondingly modest. For interventions that change the class population, reporting the engine-coupled set and its set difference alongside the percentage avoids denominator ambiguity: the percentage still describes each state, but only the set difference answers whether any component entered or left engine dependence.

5.3 Responsibility as a Decision Rule

The intervention did not apply a target architectural style, and the a-priori hypothesis of migrating to Clean Architecture was rejected during the audit, before any code was written, because the baseline already had a Unity-free domain and abstractions carrying real responsibilities.

What replaced it — assessing each abstraction against the responsibility it concretely performs — produced an outcome that no style-driven rule would have produced. It removed a DTO and four mappers, but retained three other DTOs that performed genuine representation conversion. It retained a service locator that a style-driven rule would have eliminated, because the removal attempt revealed a cross-scene lifetime responsibility that nothing else discharged. And it *introduced* two interfaces, because two concrete cycles existed for which no abstraction expressed the responsibility.

The four mechanisms identified in Section 4.1 suggest a practical ordering. Mechanisms 1 and 2 — no responsibility, and bypassed responsibility — are decidable by static inspection. Mechanisms 3 and 4 are not. Mechanism 3 (the responsibility that only the removal attempt reveals) means that static inspection can be *incomplete*: some responsibilities are visible only in the runtime composition model, which for an engine-based system lives partly in serialized data. Mechanism 4 (the justification that an earlier transformation already removed) means that a correct analysis can become obsolete during a staged intervention, which is an argument for re-deriving candidates after each stage rather than fixing the plan at the outset.

This connects to the architectural technical debt literature Verdecchia et al. (2020, 2021), which characterizes such debt as decisions that were appropriate when taken and later became obstacles. Mechanism 4 is the inverse-and-benign case: an item of apparent debt whose justification disappeared, making the item harmless rather than harmful, and detectable only by re-examination.

5.4 What Simplification Cost and What It Bought

The intervention did not buy a better metric profile. Within surviving code the indicators were flat to marginally worse, and the mean coupling proxy rose both in aggregate (+0.6) and among matched surviving types in the directly refactored group (+1.53).

The coupling increase has a concrete cause. Removing four mappers moved their transformation logic into the repositories that had called them, so those repositories now depend on more types than before. Whether this is an improvement is a design judgement, not a metric one: the system has fewer indirection layers and fewer components to keep consistent, at the cost of repositories that know more. The metric records the cost and is silent about the benefit — which is precisely the limitation that the multi-property maintainability models were designed to address Heitlager et al. (2007); Baggen et al. (2012), and a reason not to treat a coupling mean as a verdict.

Sas and Avgeriou Sas and Avgeriou (2020) found that practitioners in the embedded systems industry rarely monitor design-time qualities at all, and that trade-offs involving them therefore remain implicit. Our case is the complementary situation: the design-time indicators were monitored, closely and with a frozen instrument, and

the trade-off still did not resolve into a verdict, because the indicators price the cost of concentrating dependencies without pricing the benefit of removing an indirection layer. Monitoring is necessary for making such a trade-off explicit; on this evidence it is not sufficient for deciding it.

What the intervention did produce is structural: 56 direct accesses to engine audio singletons reduced to zero behind an existing port; two concrete cycles between views and orchestrators replaced by role contracts; one duplicated domain rule given a single owner; three dead service registrations removed. These are changes in the dependency structure that the aggregate indicators do not express, and reporting them requires the architectural narrative of Section 4.1 alongside the numbers.

We deliberately do not claim that state D is better than state D0. The study measured what changed; whether the trade is worthwhile depends on which future changes the system will actually face, and that is not observable from a static snapshot.

5.5 Implications

For *researchers*: interventions that change component population should report matched-population results alongside aggregate ones, and should separate deletion effects from restructuring effects. In this study the deletion effect was larger than the restructuring effect on every aggregate indicator.

For *practitioners* working on engine-based systems: where an intervention changes the class population, tracking the engine-coupled set alongside the percentage removes an ambiguity the percentage alone cannot resolve; and abstractions should be assessed against the responsibility they discharge in the composed, running system — not against the catalogued pattern they resemble in source Gamma et al. (1995).

For both: a retention decision, recorded with its evidence, is a result. Five of the thirteen audited concerns in this study ended in a retention decision, and two of those overturned an a-priori hypothesis. A protocol that only records changes would have reported none of it.

6 Threats to Validity

We follow the categories of Runeson and Höst Runeson and Höst (2009).

Construct validity. MI, cyclomatic complexity and our approximate efferent coupling are structural proxies, not measurements of the maintainability that developers experience. The critiques are well established: cyclomatic complexity is substantially a proxy for size Shepperd (1988), and MI is a composite calibrated on a corpus unlike ours that incorporates both Coleman et al. (1994); Ardito et al. (2020). We mitigate this by reporting the indicators separately as well as together, by reporting medians alongside means, and by never treating a movement in one of them as a statement about maintainability. For the same reason we use the term *internal software quality indicators* throughout.

Approximate efferent coupling is the weakest construct of the three. As defined in Section 3, it counts distinct capitalised identifiers within a type rather than resolved external types, so its absolute level is not a count of dependencies and should not be read as one. It both over-counts (invoked method names, properties, enum members

and constants are matched) and under-counts (dependencies expressed without a capitalised identifier are missed). We use it comparatively only: the same rule is applied to all three states, so a difference between states reflects a change in the measured text even though the level does not translate into a dependency count. Any reading of our coupling figures as C_e values would be unwarranted.

Engine dependency is defined operationally as the number of types whose declared base list contains `MonoBehaviour` or `ScriptableObject`. Two limitations follow. First, the check is *syntactic*: base types are matched as text, without resolving the inheritance graph. Second, it considers only *directly* declared base types, so a class extending an intermediate class that itself extends `MonoBehaviour` would not be counted; no such case was audited in this system, and we did not verify their absence exhaustively. The indicator therefore measures engine instantiation specifically, not engine API usage in general, and a system could reduce its use of engine APIs considerably without moving this count. Its percentage form uses classes as denominator and is sensitive to population change, which is why we report the set identity rather than the percentage as the result.

Internal validity. The eight stages were applied sequentially and may interact. The D0 checkpoint isolates dead-code removal from the architectural intervention, but it does not isolate stages B through H from one another. Two concrete interactions were in fact observed and are reported in Table 5: in both, Stage A removed the justification for a change that later stages had been expected to make. We did not measure after every stage, so finer attribution within B– H is not available.

Absence of randomisation and of a control system. This is a single-case before-and-after design. Nothing was randomised, there is no untreated comparison system evolving in parallel, and the states are successive versions of one artefact rather than independent samples. We therefore cannot exclude that some of the observed differences would have appeared under any change of comparable size, and we make no causal claim of the form that the intervention produced a given metric movement. The D0 checkpoint controls for one specific alternative explanation — dead-code removal — and for that one only. Our mitigation is to report differences as observations tied to identified source changes, and to report the mechanism for each rather than inferring it from the metric.

Concurrent structural changes. The eight stages were applied in sequence to the same tree, so the D0 $\rightarrow$ D difference aggregates all of them. A metric movement cannot be attributed to any single stage, and stages may interact — two interactions were in fact observed and are reported in Table 5. We mitigated this partially by keeping every stage in its own commit with its own gate, which supports attribution at the level of source change even where the metric cannot be decomposed; a design that measured after every stage would be stronger, and we did not do that.

Matching limitations. The matched-population analysis depends on an identity key, `file|type|member`, that does not include a parameter signature. Two overload pairs therefore collapse onto one key in each state and were excluded from matching rather than resolved by assumption. A member that was renamed or moved between files is seen as one removal plus one addition rather than as a modification, which

pushes genuine within-component change into the composition term and makes our composition effect, if anything, an over-estimate. No renames or moves of measured members occurred in this intervention, and the excluded overloads are byte-identical between D0 and D, so neither limitation affects the reported figures here; both would matter in a study with more mobile components.

External validity. This is a single case: one Unity serious game of 77 files in its final state, developed by a small academic team, with three sequentially loaded scenes. The conclusions about cross-scene lifetime depend on that scene topology. The conclusions about engine dependency should generalize to engine-based systems with a comparable composition model, but that is an expectation, not a demonstration. Nothing here supports generalization to large systems, to other engines, or to teams with different practices.

Reliability. The instrument is versioned and frozen at a single commit; each state is pinned by commit, tag and branch and recovered in an isolated worktree; determinism was confirmed by a byte-identical re-run of the whole pipeline including the analyzer output. An instrument defect found during the study was corrected, after which state C was re-measured and produced byte-identical results, establishing that all three states were measured by the same version. The measured system, the instrument, the raw per-type and per-member records and the analysis scripts are versioned in the study repository and will be published as described under Data availability.

Researcher bias. The authors performed both the intervention and its analysis, which is the standard threat in improving case studies. Partial mitigations: the classification criteria for RQ3 were fixed before measurement; the RQ3 assignment is derived mechanically from the commit history rather than from judgement after seeing the numbers; each stage passed a human approval gate before the next began; and a list of prohibited claims — including several that the data superficially appeared to support — was fixed before the manuscript was written. We did not seek to demonstrate that D is better than D0, and we report the results that contradicted our initial framing.

LLM-assisted implementation. The implementation was carried out with the assistance of an LLM coding agent under per-stage human review and approval and the recorded prohibitions (Section 3.5). The assistance may have influenced the shape of the code and the set of options considered at each stage, including surface properties such as commenting density, which is one plausible contributor to the rise in comment lines between D0 and D. The study neither measures nor claims anything about AI-assisted productivity or quality; no aspect of the design would support such a claim.

Functional equivalence. The parity evidence is strong but it is evidence, not proof. The 103 behavioural characterization tests, the clean engine import, the manual full-flow walkthrough and the post-stage smoke verification cover the behaviour they cover; the 15 architecture invariant tests attest structural properties and contribute nothing to the behavioural argument. A further limitation follows from the suite's incremental history (Table 4): only the 68 baseline tests constrained every stage, while later tests constrained only the stages that followed them. We claim behavioural preservation for that scope and no more; we do not claim formal functional equivalence. Structural differences on paths that no verification reaches were recorded rather than hidden.

7 Conclusion

This paper reported a comparative case study of responsibility-driven architectural simplification in Molz, a Unity-based serious game, measured across three pinned states of the same system with a single frozen instrument: the baseline C, a dead-code-removal control D0, and the simplified state D.

Answering RQ1, the reassessment of thirteen architectural concerns produced three removals or merges, two strengthenings, three local simplifications or changes of ownership, and five documented retentions, two of which overturned an a-priori hypothesis. The resulting system has fewer classes (65 to 60) but more interfaces (12 to 14) and more members (451 to 457): simplification here meant removing abstractions that carried no responsibility while introducing two that expressed one.

Answering RQ2, the aggregate indicators moved very little between D0 and D — MI 71.7 to 71.8, CC 2.39 to 2.38, coupling proxy 12.8 to 13.4 — and a matched-population analysis over the 442 members present in both states reversed the direction of the first two: population composition determined the direction the aggregate reported, while the surviving members moved slightly the other way. All four magnitudes are small in absolute terms, and we infer no practical significance from them; the result of interest is the sensitivity of the aggregate to population composition, not the direction of a hundredth of a point. The set of engine-coupled types was identical in both states; only its percentage moved, through denominator change.

Answering RQ3, the measured changes were contained outside the source-preserved group, whose 64 types remained metric-identical across the intervention, and fell within the directly and indirectly affected groups. Because the preserved group is defined by the absence of relevant source changes, this containment is consistent with the classification rather than an independent confirmation of it. The dead code removed in the control had higher MI and lower cyclomatic complexity and a lower coupling proxy than the system average (MI 80.3 against 72.5; CC 1.78 against 2.33; coupling 4.50 against 11.4), so removing it shifted the composition of the measured population and moved every normalized aggregate mean in the less favourable direction — an effect that, without the control state, would have been credited to the architectural intervention.

The study's central methodological claim is that interventions which change the population of components cannot be evaluated by aggregate means alone. Its central empirical observation is narrower and strictly local to this case: in this intervention, the set of engine-dependent types remained unchanged despite the layer-level reorganization. We do not generalize that to engine-based systems at large, and Section 5 notes that a differently targeted intervention might well have moved the set.

Several directions remain open. Measuring after each stage rather than only at the endpoints would permit attribution within the intervention. Repeating the design on systems built with other engines, and on systems large enough for the population effects to be studied statistically, would test how far the observations extend. A companion study of the analyzer findings collected here, which this paper deliberately leaves uninterpreted, is a separate line of work. Finally, the responsibility-driven decision rule was applied by the authors; how consistently different engineers would apply it to the same system is an empirical question we did not address.

Acknowledgements. This work was carried out within an undergraduate research programme at the Universidade Federal do Agreste de Pernambuco. The authors thank the members of the Molz project team for their contributions to the system studied here.

Declarations

Funding. NEEDS SOURCE — funding programme and grant identifiers to be supplied.

Competing interests. The authors declare no competing interests.

Ethics approval and consent to participate. Not applicable. This study analyses source code artefacts and involves no human participants.

Use of generative AI. An LLM-based coding agent (Claude Code, Anthropic) assisted the implementation of the architectural intervention, under the staged human-approval protocol documented in Section 3.5 and discussed as a threat in Section 6. Generative AI was not used to create or alter the measurement records.

Data availability. The research data supporting this study comprise the frozen per-type and per-member measurements for states C, D0 and D; the sanitized static-analysis outputs; the RQ3 component classification; the test-execution records; and the state-freeze and results-analysis records used to derive the reported comparisons. These materials have been assembled in the study replication package and will be deposited in Zenodo before submission. Persistent identifier: [**ZENODO DOI PENDING**]. The source repository for the system under study is publicly accessible, and state C is available there (https://github.com/bcccoworkingufape/game-quimica-2D).

Code availability. The code required to reproduce the reported analyses comprises the frozen measurement instrument, the aggregation and matched-population analysis scripts, the characterization and architecture-invariant tests, the environment manifest, and source snapshots of states C, D0 and D containing the project-owned code and research data required for replication. These materials are included in the same replication package and will be deposited in Zenodo before submission. Persistent identifier: [**ZENODO DOI PENDING**]. Reproduction instructions are provided in the accompanying README. Third-party Unity assets and packages are not redistributed in the replication package; applicable licence terms will be stated in the Zenodo record.

Author contributions. NEEDS SOURCE — CRediT contribution statements to be supplied.

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